

# Addressing Questions/Comments received for DY absolute Cross-Section Study

Chatura Kuruppu

New Mexico State University

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This talk addresses additional comments received for the talk DocID: 11584  
(<https://sequest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11584>)

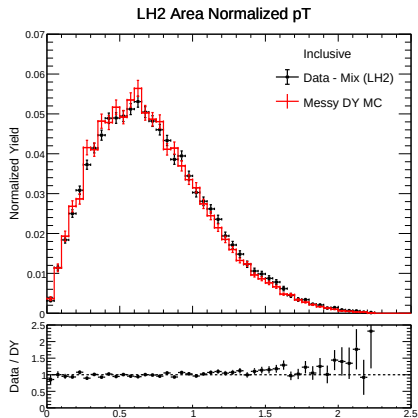
## Supplemental Documents:

- Technote  
(<https://sequest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11569>)
- Previous talk  
(<https://sequest-docdb.fnal.gov/cgi-bin/sso/ShowDocument?docid=11584>)

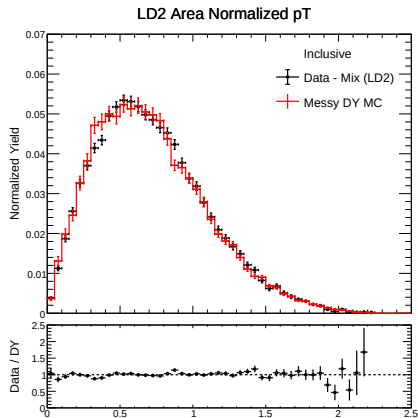
*“The other comparison that I raised on Friday was of the mass and  $x_F$  distributions in each  $p_T$  bin. Can you make comparison plots? As you integrate the events over mass and  $x_F$  when deriving  $d\sigma/dp_T$ , the agreement between the real data and the MC is crucial”*

# $p_T$ Inclusive (Data - Mix): LH2 vs LD2

## LH2 Target



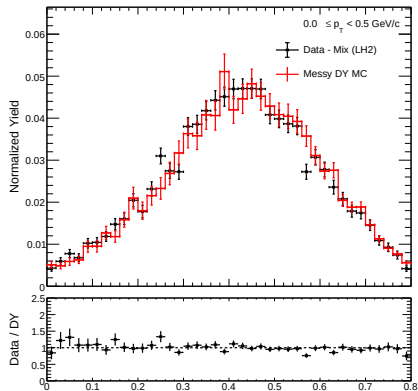
## LD2 Target



# $x_F$ ( $0.0 \leq p_T < 0.5$ GeV/c): LH2 vs LD2

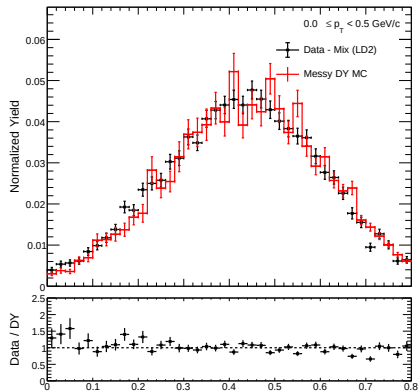
## LH2 Target

LH2 Area Normalized  $x_F$



## LD2 Target

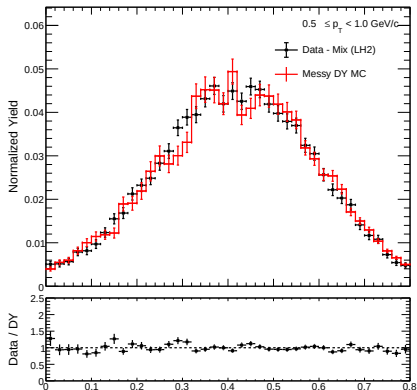
LD2 Area Normalized  $x_F$



# $x_F$ ( $0.5 \leq p_T < 1.0$ GeV/c): LH2 vs LD2

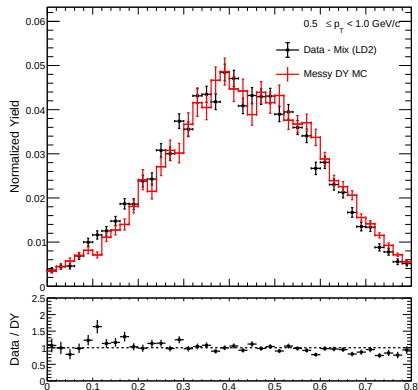
## LH2 Target

LH2 Area Normalized  $x_F$



## LD2 Target

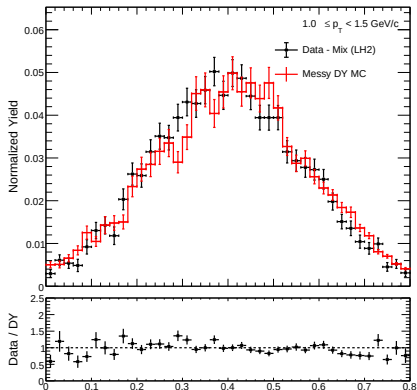
LD2 Area Normalized  $x_F$



# $x_F$ ( $1.0 \leq p_T < 1.5$ GeV/c): LH2 vs LD2

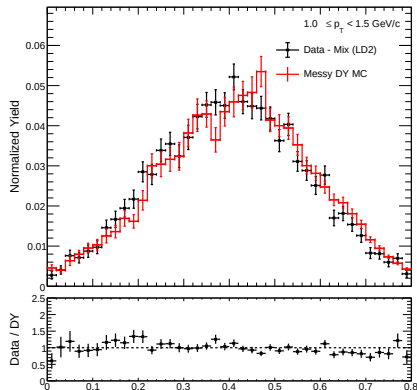
## LH2 Target

LH2 Area Normalized  $x_F$



## LD2 Target

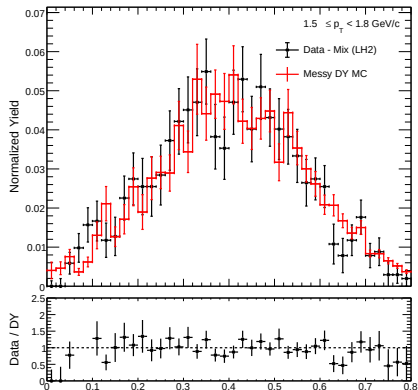
LD2 Area Normalized  $x_F$



# $x_F$ ( $1.5 \leq p_T < 1.8$ GeV/c): LH2 vs LD2

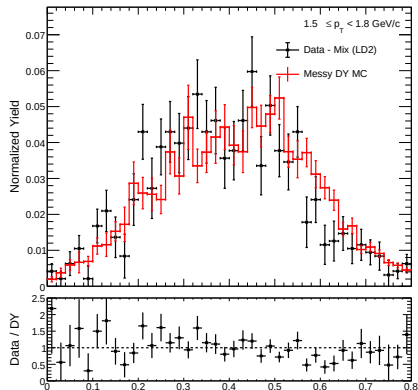
## LH2 Target

LH2 Area Normalized  $x_F$



## LD2 Target

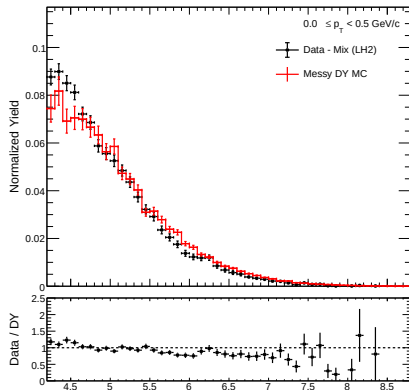
LD2 Area Normalized  $x_F$



# Mass ( $0.0 \leq p_T < 0.5$ GeV/c): LH2 vs LD2

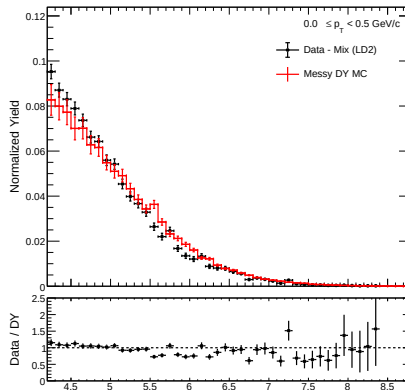
## LH2 Target

LH2 Area Normalized Mass



## LD2 Target

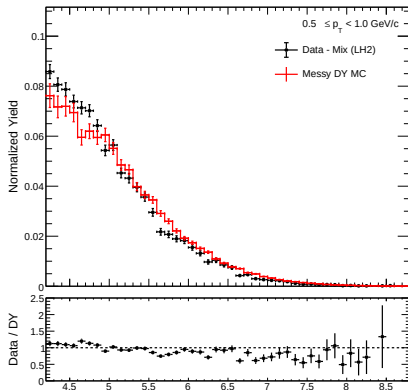
LD2 Area Normalized Mass



# Mass ( $0.5 \leq p_T < 1.0$ GeV/c): LH2 vs LD2

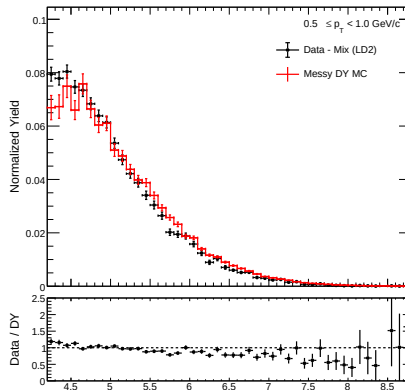
## LH2 Target

LH2 Area Normalized Mass



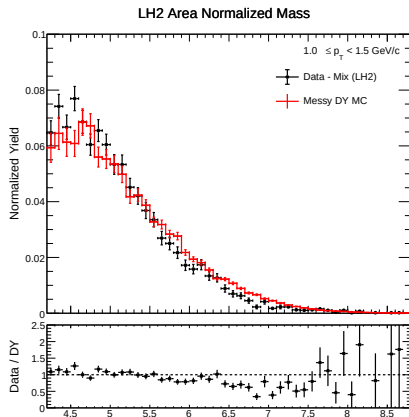
## LD2 Target

LD2 Area Normalized Mass

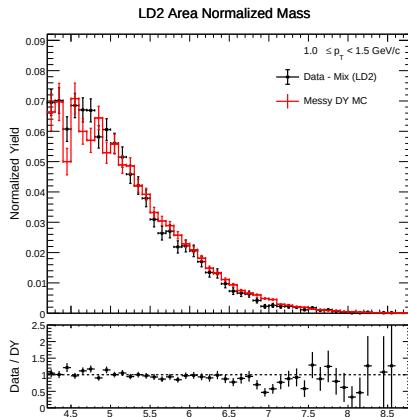


# Mass ( $1.0 \leq p_T < 1.5$ GeV/c): LH2 vs LD2

## LH2 Target



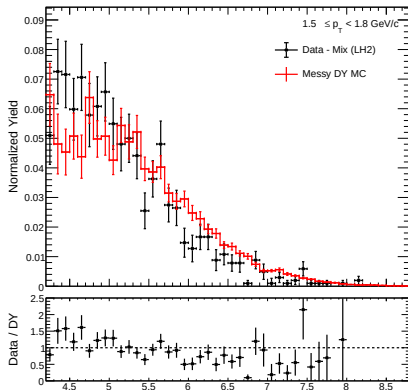
## LD2 Target



# Mass ( $1.5 \leq p_T < 1.8$ GeV/c): LH2 vs LD2

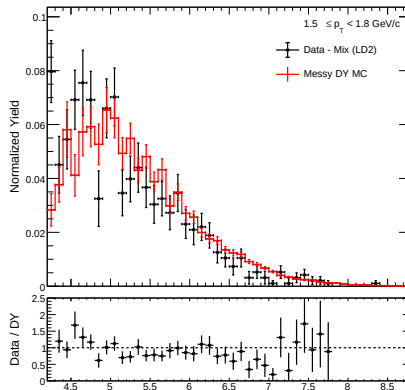
## LH2 Target

LH2 Area Normalized Mass



## LD2 Target

LD2 Area Normalized Mass



# Systematic Correlations to cross-section ratio

*“How was the systematic error propagated from the LH2 and LD2 results? Have you considered any correlation?”*

# The Observable: Cross-Section Ratio

**Objective:** Calculate the error for the single differential cross-section ratio in a given kinematic bin (e.g.,  $p_T$ ).

The ratio  $R$  is defined as:

$$R = \frac{\sigma_{pd}}{2\sigma_{pp}} \equiv \frac{A}{cB} \quad (1)$$

Where:

- $A = \sigma_{pd}$  (Liquid Deuterium cross-section)
- $B = \sigma_{pp}$  (Liquid Hydrogen cross-section)
- $c = 2$  (Exact scaling constant, zero uncertainty)

# General Error Propagation (The Master Equation)

For any function  $R(A, B)$ , the variance is given by the partial derivatives and the covariance matrix:

$$\delta R^2 = \left(\frac{\partial R}{\partial A}\right)^2 \delta A^2 + \left(\frac{\partial R}{\partial B}\right)^2 \delta B^2 + 2 \left(\frac{\partial R}{\partial A}\right) \left(\frac{\partial R}{\partial B}\right) \text{cov}(A, B) \quad (2)$$

By defining the covariance in terms of the Pearson correlation coefficient  $\rho$ :

$$\text{cov}(A, B) = \rho(\delta A)(\delta B) \quad \text{where } \rho \in [-1, 1] \quad (3)$$

Evaluating the derivatives for  $R = A/(cB)$  and converting to relative errors yields the master equation:

$$\left(\frac{\delta R}{R}\right)^2 = \left(\frac{\delta A}{A}\right)^2 + \left(\frac{\delta B}{B}\right)^2 - 2\rho \left(\frac{\delta A}{A}\right) \left(\frac{\delta B}{B}\right) \quad (4)$$

# 1. Statistical Uncertainties ( $\rho = 0$ )

**Physical Reality:** Statistical fluctuations are driven by random counting statistics in finite data samples.

A random upward fluctuation in the  $pd$  dataset has no influence on the random fluctuations in the  $pp$  dataset. Therefore, they are **completely uncorrelated** ( $\rho = 0$ ).

Substituting  $\rho = 0$  into the master equation:

$$\left(\frac{\delta R_{\text{stat}}}{R}\right)^2 = \left(\frac{\delta\sigma_{pd}^{\text{stat}}}{\sigma_{pd}}\right)^2 + \left(\frac{\delta\sigma_{pp}^{\text{stat}}}{\sigma_{pp}}\right)^2 - 2(0)\left(\frac{\delta\sigma_{pd}^{\text{stat}}}{\sigma_{pd}}\right)\left(\frac{\delta\sigma_{pp}^{\text{stat}}}{\sigma_{pp}}\right)$$

The covariance term vanishes, yielding the quadrature sum:

$$\frac{\delta R_{\text{stat}}}{R} = \sqrt{\left(\frac{\delta\sigma_{pd}^{\text{stat}}}{\sigma_{pd}}\right)^2 + \left(\frac{\delta\sigma_{pp}^{\text{stat}}}{\sigma_{pp}}\right)^2} \quad (5)$$

## 2. Systematic Uncertainties ( $\rho = 1$ )

**Physical Reality:** Systematic biases (e.g., tracking efficiency, geometric acceptance) apply to the exact same spectrometer running under identical beam conditions.

If an inefficiency causes us to systematically underestimate  $\sigma_{pd}$ , it will also cause us to systematically underestimate  $\sigma_{pp}$ . They move together (**perfectly positively correlated**,  $\rho = 1$ ).

Substituting  $\rho = 1$  into the master equation:

$$\left(\frac{\delta R_{\text{sys}}}{R}\right)^2 = \left(\frac{\delta\sigma_{pd}^{\text{sys}}}{\sigma_{pd}}\right)^2 + \left(\frac{\delta\sigma_{pp}^{\text{sys}}}{\sigma_{pp}}\right)^2 - 2(1)\left(\frac{\delta\sigma_{pd}^{\text{sys}}}{\sigma_{pd}}\right)\left(\frac{\delta\sigma_{pp}^{\text{sys}}}{\sigma_{pp}}\right)$$

This forms a perfect square  $x^2 + y^2 - 2xy = (x - y)^2$ :

$$\left(\frac{\delta R_{\text{sys}}}{R}\right)^2 = \left(\frac{\delta\sigma_{pd}^{\text{sys}}}{\sigma_{pd}} - \frac{\delta\sigma_{pp}^{\text{sys}}}{\sigma_{pp}}\right)^2 \quad (6)$$

# The Power of Ratios: Error Cancellation

Taking the square root gives our final systematic error propagation:

$$\frac{\delta R_{\text{sys}}}{R} = \left| \frac{\delta \sigma_{pd}^{\text{sys}}}{\sigma_{pd}} - \frac{\delta \sigma_{pp}^{\text{sys}}}{\sigma_{pp}} \right| \quad (7)$$

## Why this matters:

- If systematic errors are evaluated as uncorrelated, they inflate the final uncertainty (quadrature sum).
- By correctly identifying them as perfectly correlated ( $\rho = 1$ ), identical fractional biases subtract and **cancel out entirely**.
- This mathematically isolates the physical cross-section ratio from global detector biases.

# Total Bin-by-Bin Uncertainty

Finally, we evaluate the total error for each kinematic bin.

Because the sources of statistical scatter (random sample size) and systematic bias (global normalization/efficiencies) are independent of each other, their cross-correlation is zero.

We add the absolute propagated errors in quadrature:

$$\delta R_{\text{total}} = \sqrt{(\delta R_{\text{stat}})^2 + (\delta R_{\text{sys}})^2} \quad (8)$$

This  $\delta R_{\text{total}}$  represents the final error bars plotted on the single differential cross-section ratio  $d\sigma_{pd}/2d\sigma_{pp}$  vs  $p_T$ .

# Error Table Comparison

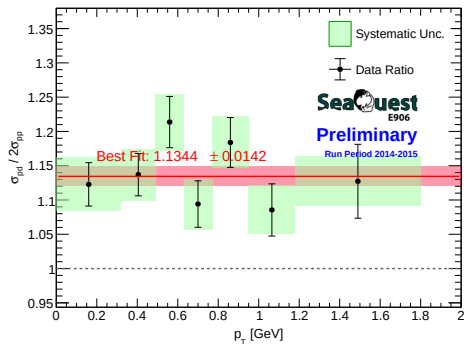
**Summary Table:** Extracted bin uncertainties comparing the uncorrelated background treatment against the new 100% correlated systematic treatment.

| Uncorrelated (Old) |        |          |         |           |
|--------------------|--------|----------|---------|-----------|
| $p_T$ Bin [GeV]    | Ratio  | Stat Err | Sys Err | Total Err |
| [0.00, 0.32)       | 1.1228 | 0.0316   | 0.0392  | 0.0504    |
| [0.32, 0.49)       | 1.1370 | 0.0309   | 0.0376  | 0.0487    |
| [0.49, 0.63)       | 1.2137 | 0.0374   | 0.0413  | 0.0557    |
| [0.63, 0.77)       | 1.0941 | 0.0339   | 0.0365  | 0.0498    |
| [0.77, 0.95)       | 1.1839 | 0.0364   | 0.0377  | 0.0524    |
| [0.95, 1.18)       | 1.0855 | 0.0381   | 0.0351  | 0.0518    |
| [1.18, 1.80)       | 1.1273 | 0.0539   | 0.0362  | 0.0649    |

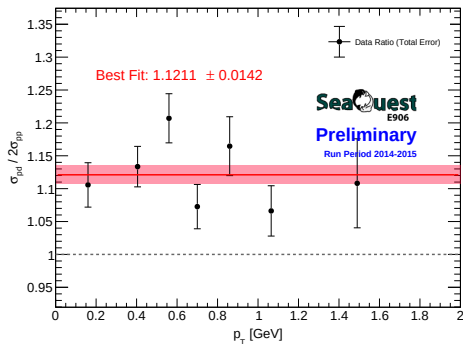
| Correlated (Latest) |        |          |         |           |
|---------------------|--------|----------|---------|-----------|
| $p_T$ Bin [GeV]     | Ratio  | Stat Err | Sys Err | Total Err |
| [0.00, 0.32)        | 1.1057 | 0.0311   | 0.0130  | 0.0338    |
| [0.32, 0.49)        | 1.1335 | 0.0308   | 0.0006  | 0.0308    |
| [0.49, 0.63)        | 1.2070 | 0.0372   | 0.0037  | 0.0374    |
| [0.63, 0.77)        | 1.0727 | 0.0332   | 0.0056  | 0.0337    |
| [0.77, 0.95)        | 1.1646 | 0.0358   | 0.0269  | 0.0448    |
| [0.95, 1.18)        | 1.0662 | 0.0374   | 0.0080  | 0.0383    |
| [1.18, 1.80)        | 1.1081 | 0.0530   | 0.0421  | 0.0677    |

# Ratio Plot Comparison

## Uncorrelated Background Treatment (Old)



## 100% Correlated Propagation (Latest)



# Weighted Average Cross-Section

*“Can you compute the weighted standard deviation of the cross sections of RS 57, 59, 62, 67 and 70? It should be better in the sense that we treat the roadsets equally, since we cannot assure RS 67 is 100% correct.”*

**This requires collaboration input!**

# Variance Calculation Mathematics

For a given kinematic bin, let  $x_i$  be the absolute cross-section for roadset  $i$ , and  $e_i$  be its statistical error.

## Inverse-Variance Weighting:

To suppress massive Poisson fluctuations from low-POT datasets (e.g., RS59), we scale by statistical power ( $w_i = 1/e_i^2$ ):

$$\mu_w = \frac{\sum w_i x_i}{\sum w_i}$$

$$\sigma_{\text{sys},w} = \sqrt{\frac{\sum w_i}{(\sum w_i)^2 - \sum w_i^2} \sum w_i (x_i - \mu_w)^2}$$

## Resulting Uncertainties (1D $p_T$ - LH2)

Using the inverse-variance weighted strategy suppresses low-statistics noise, yielding the following systematic errors for Liquid Hydrogen.

*Note: Bins with zero relative error or relative error  $\geq 50\%$  have been omitted.*

| $p_T$ Bin | Range [GeV]  | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|-----------|--------------|-----------------|--------------------------|----------------|
| 0         | [0.00, 0.32) | 0.001794        | 0.000290                 | 16.18%         |
| 1         | [0.32, 0.49) | 0.003872        | 0.000511                 | 13.19%         |
| 2         | [0.49, 0.63) | 0.004383        | 0.000536                 | 12.24%         |
| 3         | [0.63, 0.77) | 0.004766        | 0.000651                 | 13.65%         |
| 4         | [0.77, 0.95) | 0.004313        | 0.000533                 | 12.37%         |
| 5         | [0.95, 1.18) | 0.003721        | 0.000662                 | 17.79%         |
| 6         | [1.18, 1.80) | 0.001853        | 0.000248                 | 13.40%         |

## Resulting Uncertainties (1D $p_T$ - LD2)

Applying the same inverse-variance weighted procedure to Liquid Deuterium targets yields comparable systematic margins.

*Note: Bins with zero relative error or relative error  $\geq 50\%$  have been omitted.*

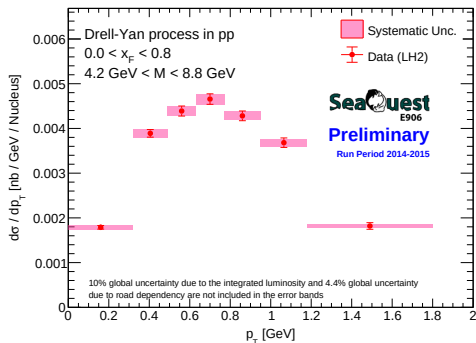
| $p_T$ Bin | Range [GeV]  | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|-----------|--------------|-----------------|--------------------------|----------------|
| 0         | [0.00, 0.32) | 0.003968        | 0.000698                 | 17.59%         |
| 1         | [0.32, 0.49) | 0.008777        | 0.001154                 | 13.15%         |
| 2         | [0.49, 0.63) | 0.010581        | 0.001340                 | 12.67%         |
| 3         | [0.63, 0.77) | 0.010224        | 0.001331                 | 13.02%         |
| 4         | [0.77, 0.95) | 0.010045        | 0.001534                 | 15.27%         |
| 5         | [0.95, 1.18) | 0.007935        | 0.001482                 | 18.67%         |
| 6         | [1.18, 1.80) | 0.004107        | 0.000738                 | 17.98%         |

# List of systematics

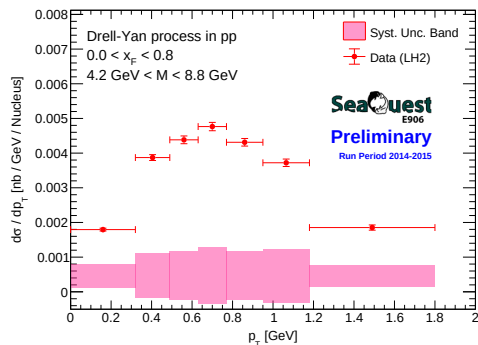
- Acceptance Correction
- Efficiency Correction for Yields
- $\psi'$  Contamination
- Road Dependency

# LH2 Cross-Section Previous Vs Latest with updated systematics

## Previous

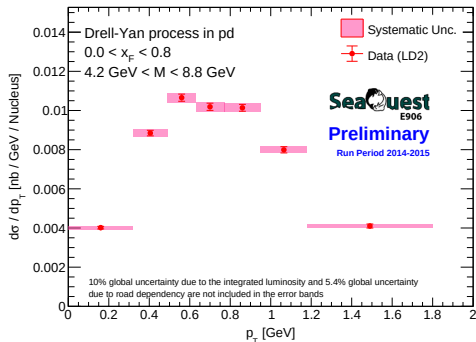


## Latest (with weighted average)

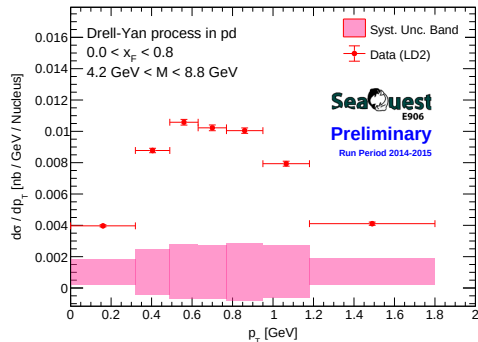


# LD2 Cross-Section Previous Vs Latest with updated systematics

## Previous

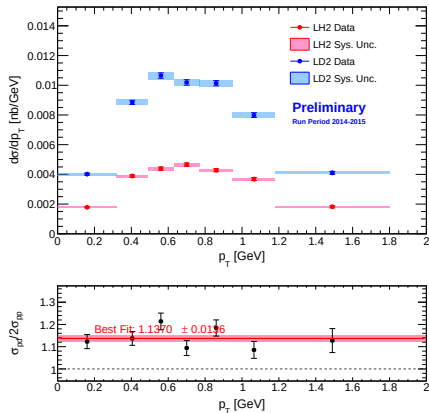


## Latest (with weighted average)

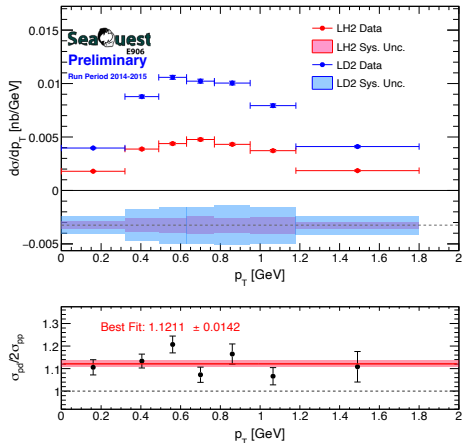


# Summary Plots for $p_T$ Previous Vs Latest with updated systematics

## Previous

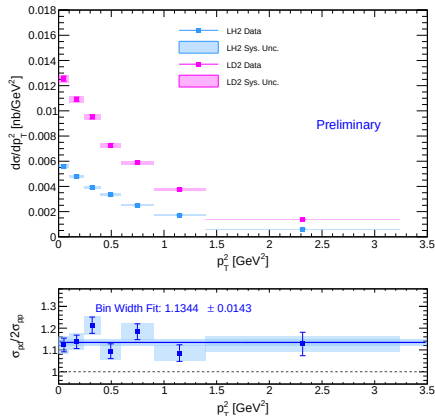


## Latest (with weighted average)

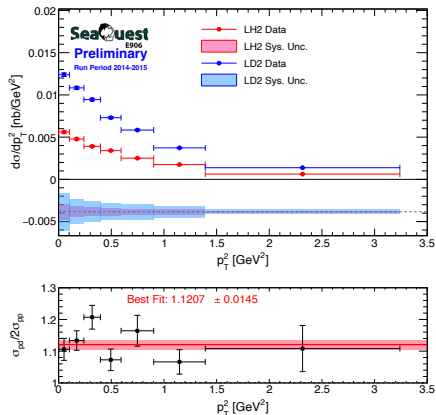


# Summary Plots for $p_T^2$ Previous Vs Latest with updated systematics

## Previous

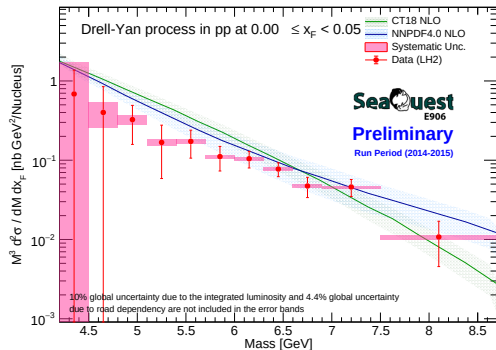


## Latest (with weighted average)

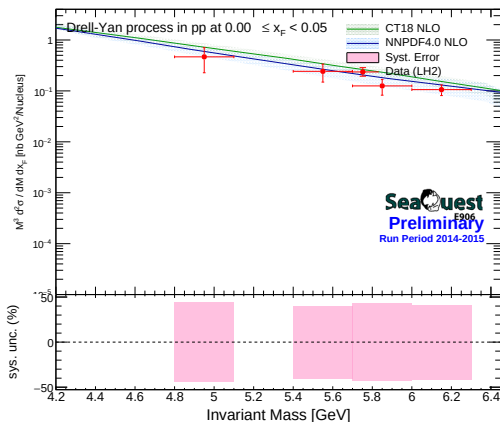


# LH2 Cross-Section Previous Vs Latest: $0.00 \leq x_F < 0.05$

## Previous

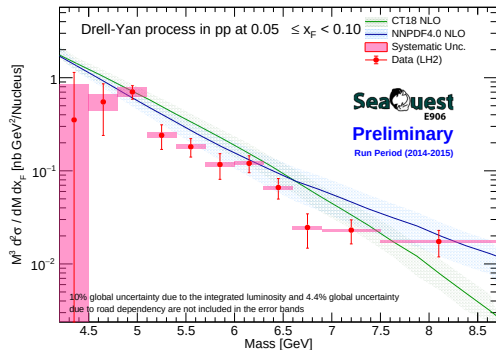


## Latest (with weighted average)

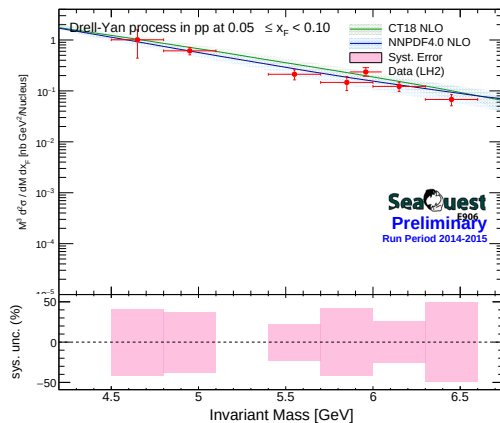


# LH2 Cross-Section Previous Vs Latest: $0.05 \leq x_F < 0.10$

## Previous

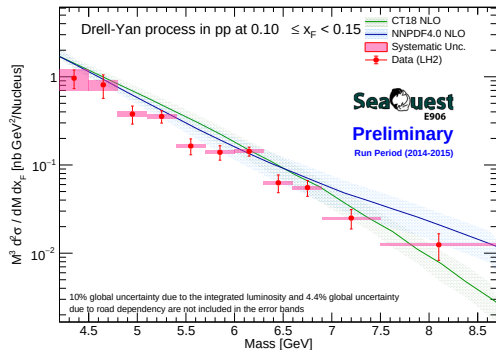


## Latest (with weighted average)

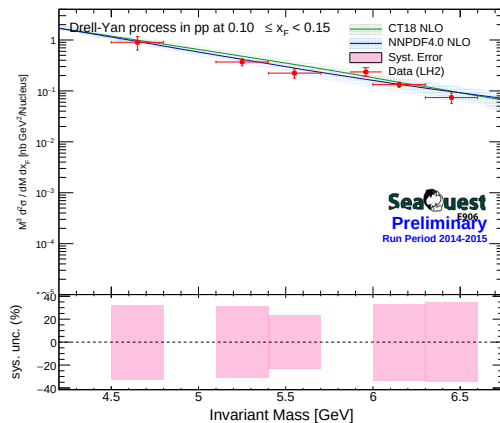


# LH2 Cross-Section Previous Vs Latest: $0.10 \leq x_F < 0.15$

## Previous

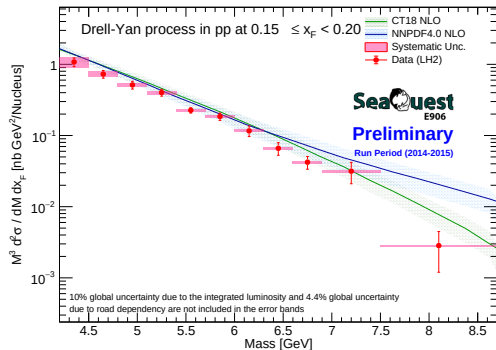


## Latest (with weighted average)

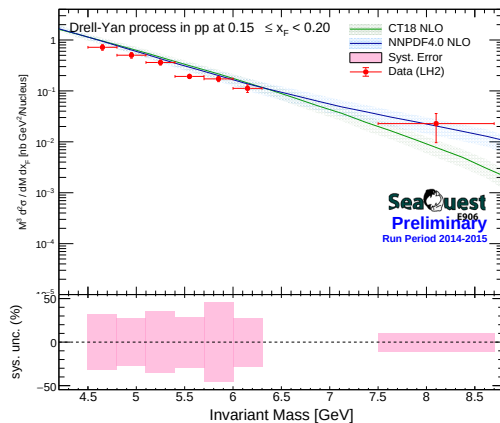


# LH2 Cross-Section Previous Vs Latest: $0.15 \leq x_F < 0.20$

## Previous

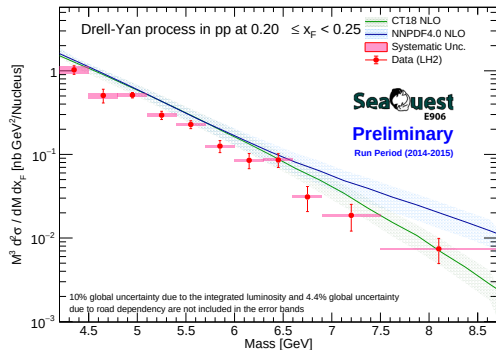


## Latest (with weighted average)

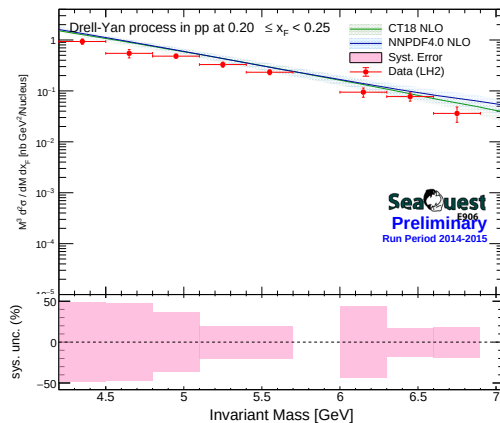


# LH2 Cross-Section Previous Vs Latest: $0.20 \leq x_F < 0.25$

## Previous

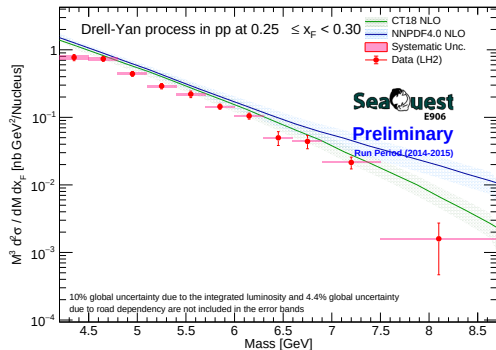


## Latest (with weighted average)

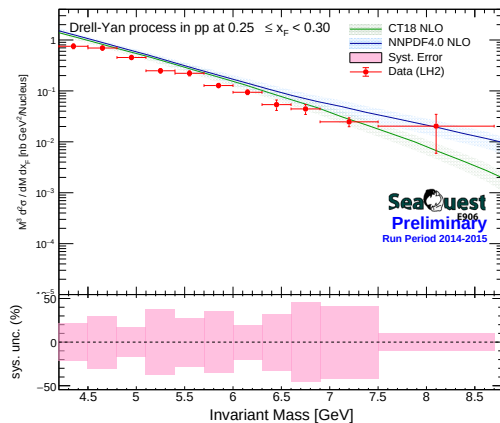


# LH2 Cross-Section Previous Vs Latest: $0.25 \leq x_F < 0.30$

## Previous

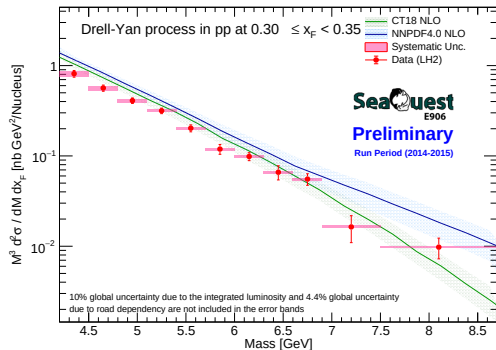


## Latest (with weighted average)

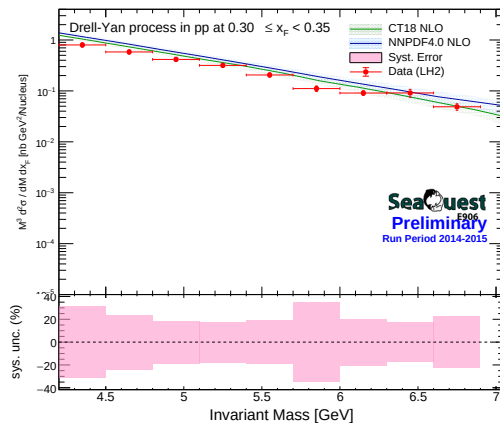


# LH2 Cross-Section Previous Vs Latest: $0.30 \leq x_F < 0.35$

## Previous

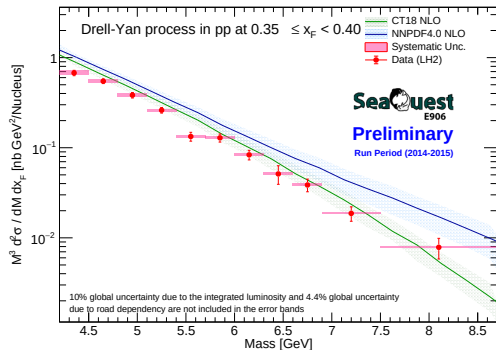


## Latest (with weighted average)

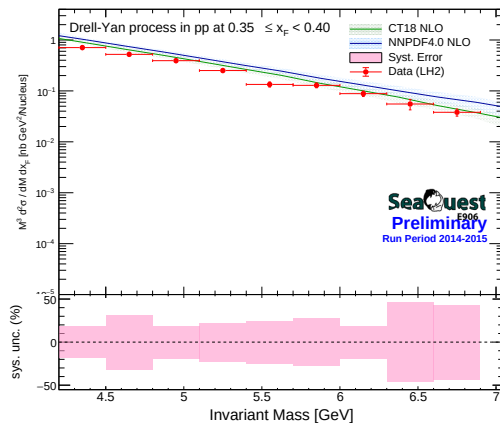


# LH2 Cross-Section Previous Vs Latest: $0.35 \leq x_F < 0.40$

## Previous

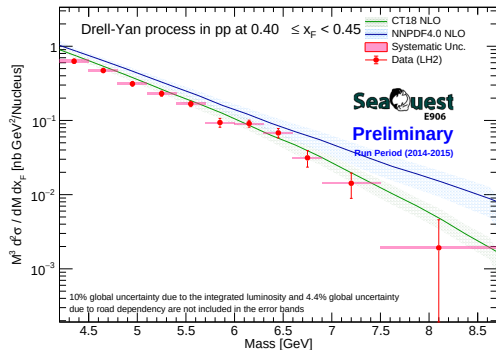


## Latest (with weighted average)

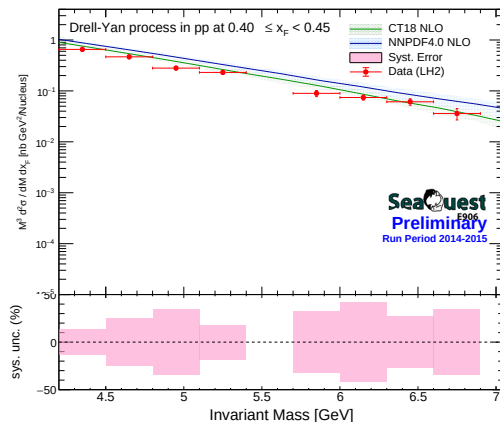


# LH2 Cross-Section Previous Vs Latest: $0.40 \leq x_F < 0.45$

## Previous

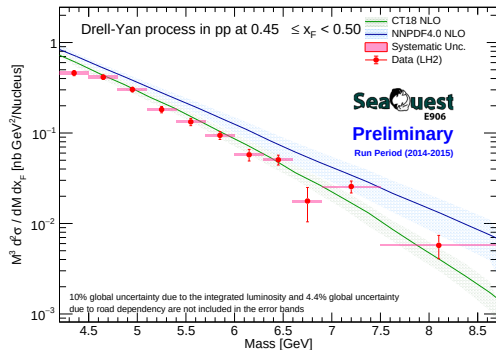


## Latest (with weighted average)

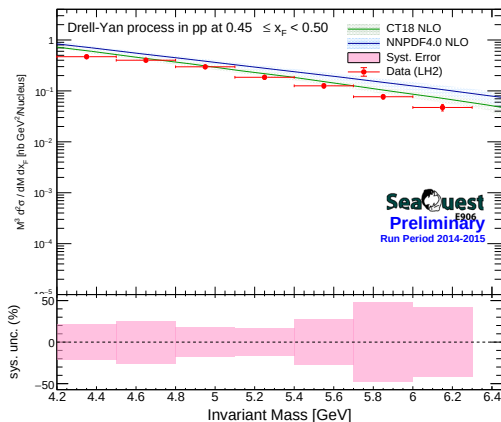


# LH2 Cross-Section Previous Vs Latest: $0.45 \leq x_F < 0.50$

## Previous

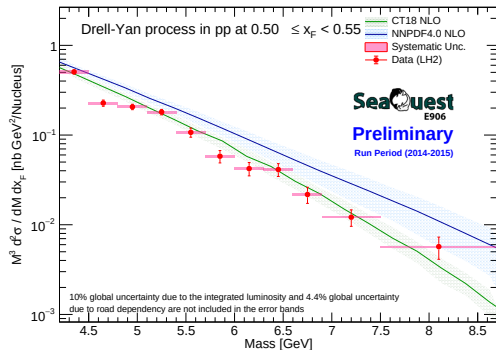


## Latest (with weighted average)

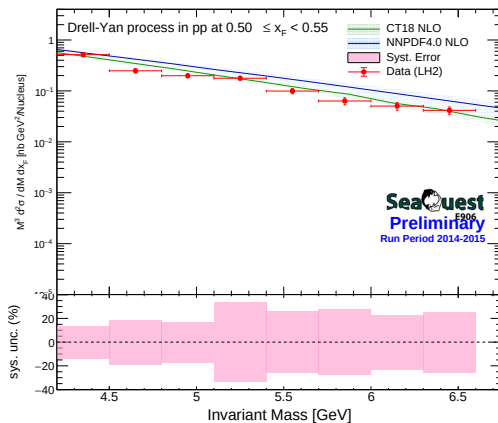


# LH2 Cross-Section Previous Vs Latest: $0.50 \leq x_F < 0.55$

## Previous

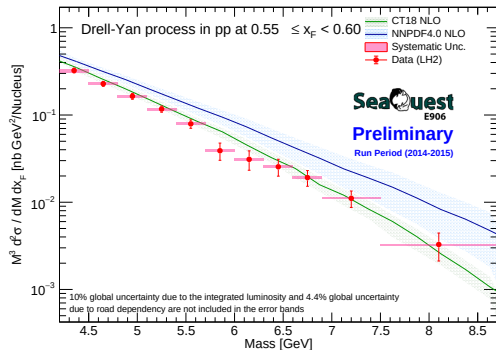


## Latest (with weighted average)

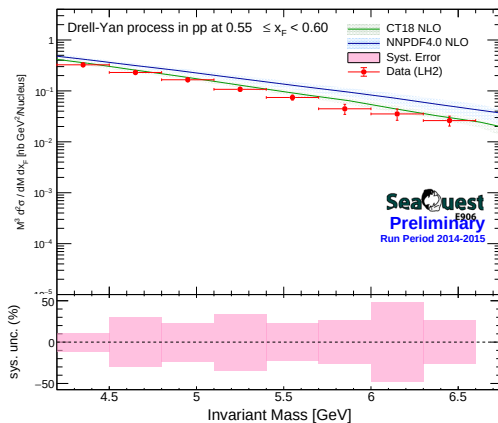


# LH2 Cross-Section Previous Vs Latest: $0.55 \leq x_F < 0.60$

## Previous

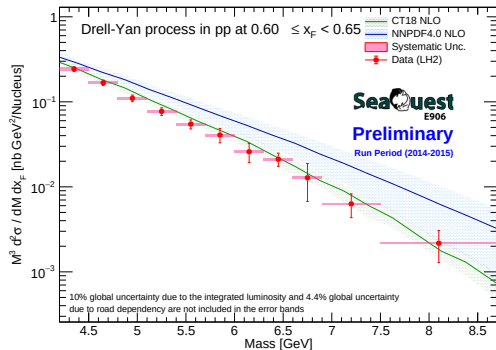


## Latest (with weighted average)

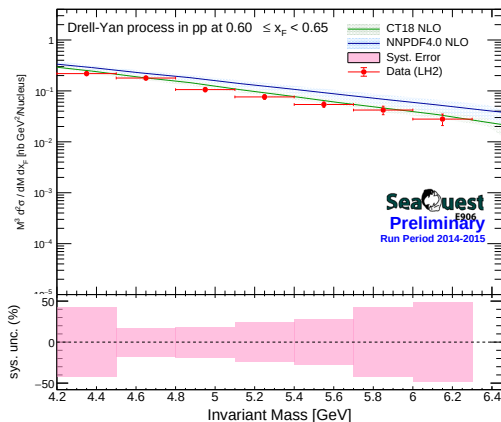


# LH2 Cross-Section Previous Vs Latest: $0.60 \leq x_F < 0.65$

## Previous

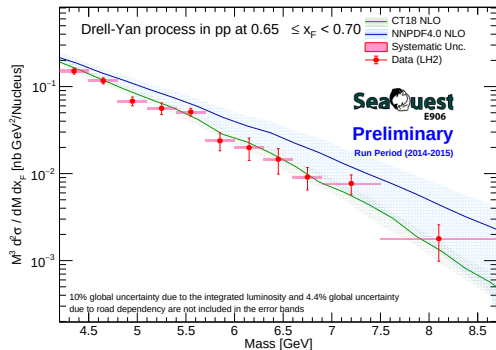


## Latest (with weighted average)

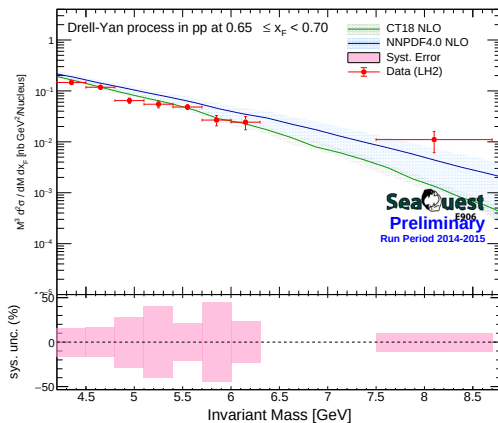


# LH2 Cross-Section Previous Vs Latest: $0.65 \leq x_F < 0.70$

## Previous

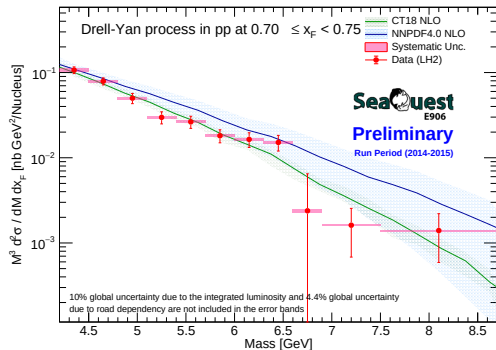


## Latest (with weighted average)

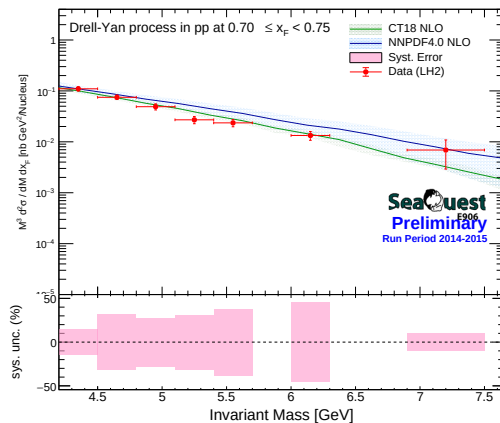


# LH2 Cross-Section Previous Vs Latest: $0.70 \leq x_F < 0.75$

## Previous

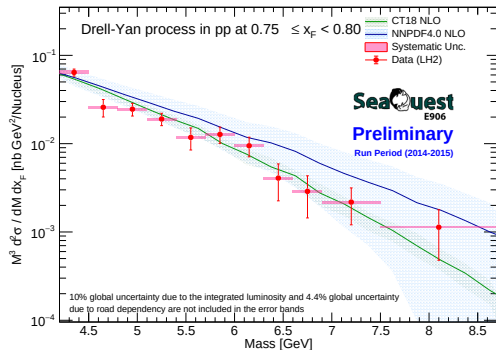


## Latest (with weighted average)

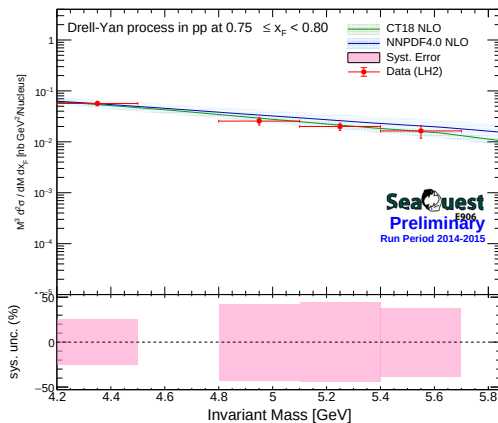


# LH2 Cross-Section Previous Vs Latest: $0.75 \leq x_F < 0.80$

## Previous

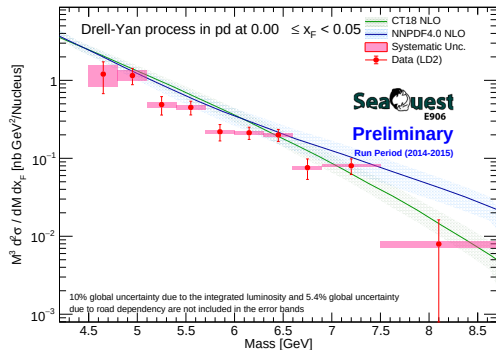


## Latest (with weighted average)

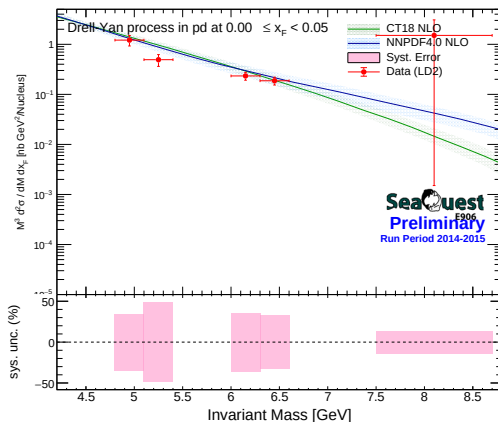


# LD2 Cross-Section Previous Vs Latest: $0.00 \leq x_F < 0.05$

## Previous

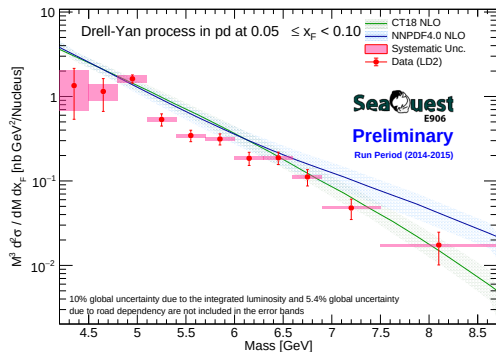


## Latest (with weighted average)

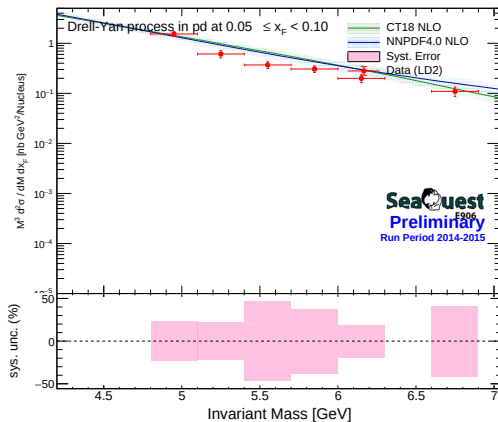


# LD2 Cross-Section Previous Vs Latest: $0.05 \leq x_F < 0.10$

## Previous

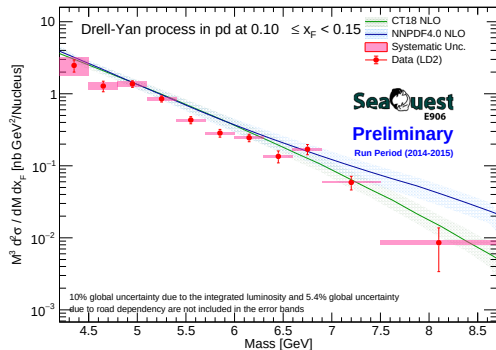


## Latest (with weighted average)

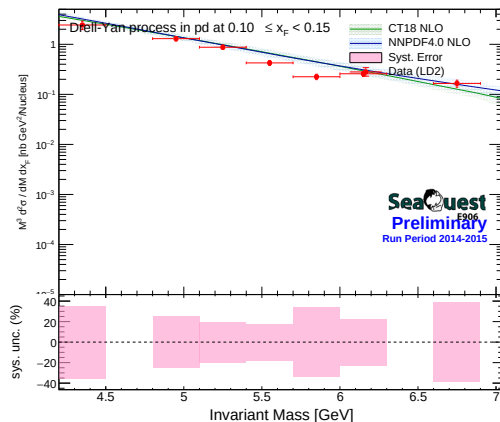


# LD2 Cross-Section Previous Vs Latest: $0.10 \leq x_F < 0.15$

## Previous

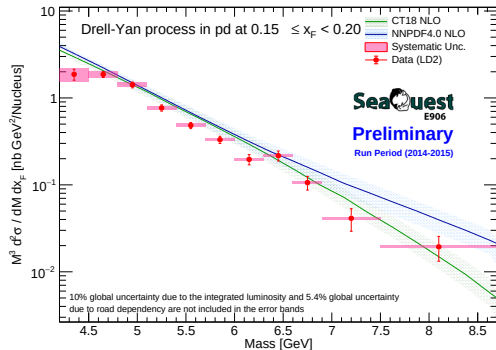


## Latest (with weighted average)

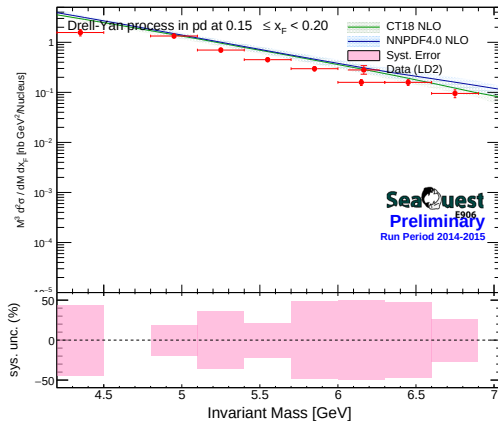


# LD2 Cross-Section Previous Vs Latest: $0.15 \leq x_F < 0.20$

## Previous

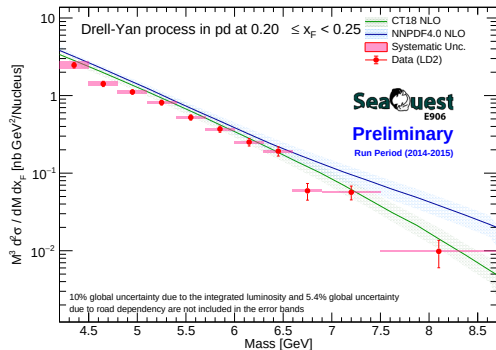


## Latest (with weighted average)

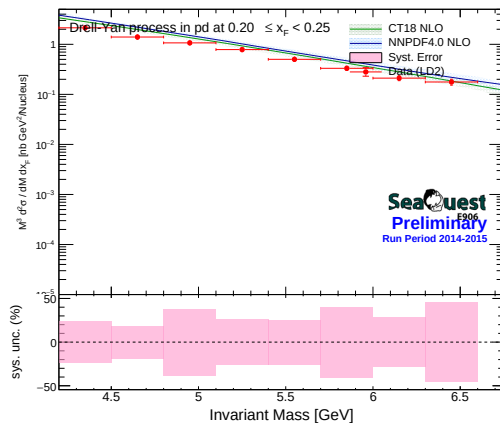


# LD2 Cross-Section Previous Vs Latest: $0.20 \leq x_F < 0.25$

## Previous

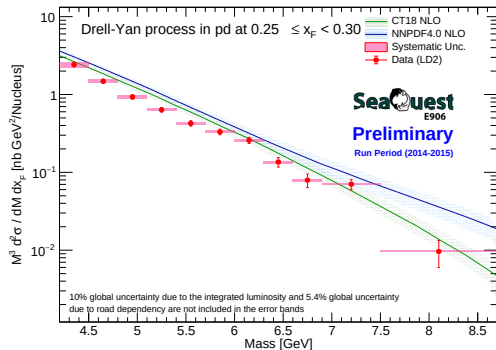


## Latest (with weighted average)

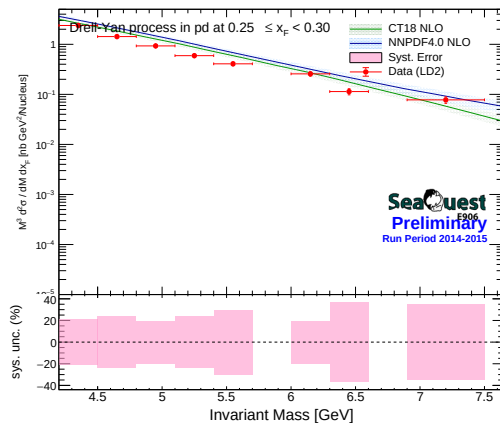


# LD2 Cross-Section Previous Vs Latest: $0.25 \leq x_F < 0.30$

## Previous

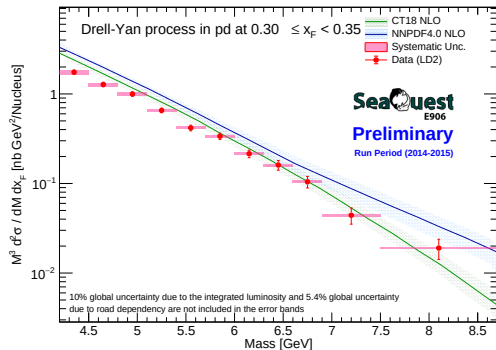


## Latest (with weighted average)

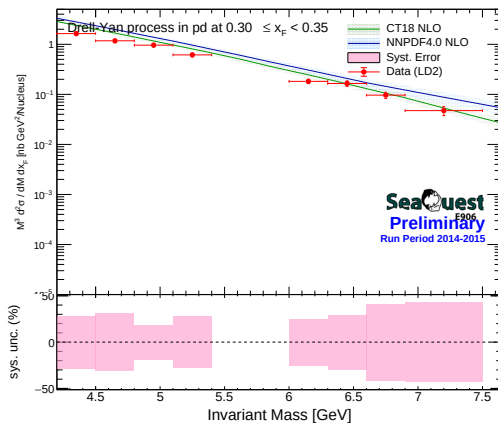


# LD2 Cross-Section Previous Vs Latest: $0.30 \leq x_F < 0.35$

## Previous

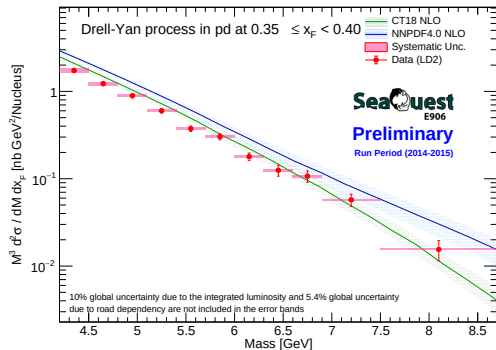


## Latest (with weighted average)

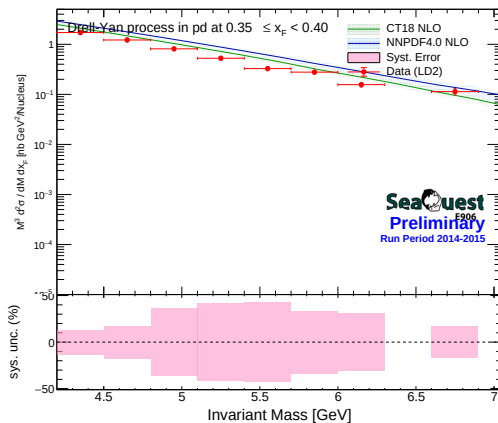


# LD2 Cross-Section Previous Vs Latest: $0.35 \leq x_F < 0.40$

## Previous

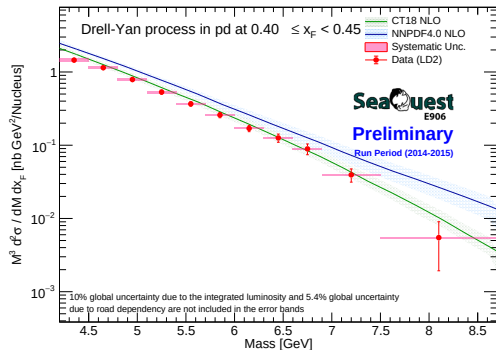


## Latest (with weighted average)

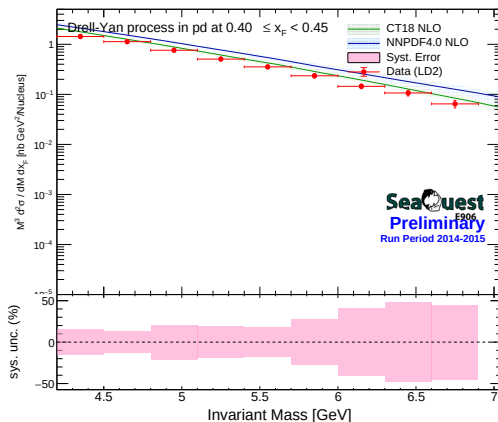


# LD2 Cross-Section Previous Vs Latest: $0.40 \leq x_F < 0.45$

## Previous

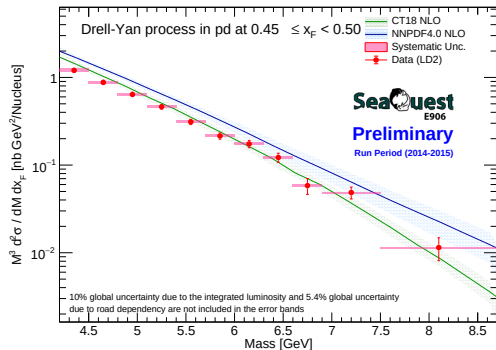


## Latest (with weighted average)

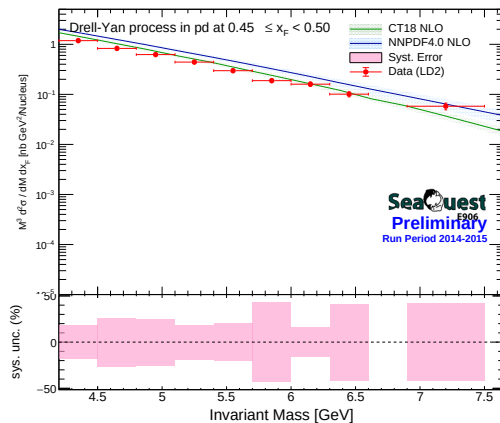


# LD2 Cross-Section Previous Vs Latest: $0.45 \leq x_F < 0.50$

## Previous

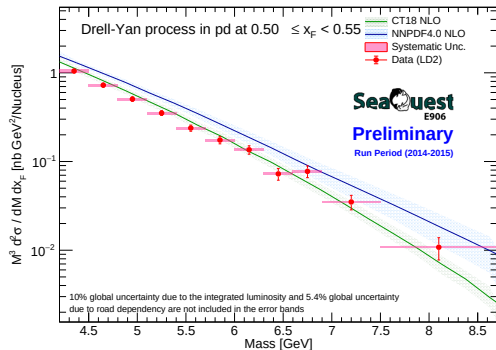


## Latest (with weighted average)

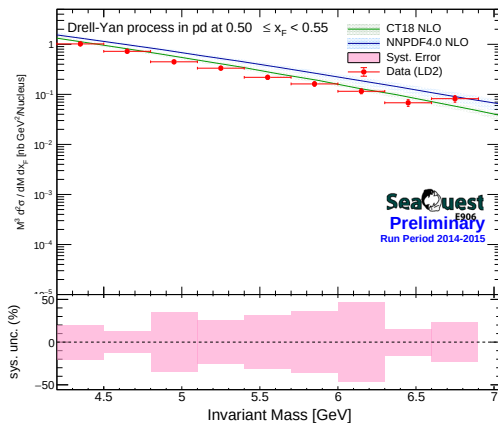


# LD2 Cross-Section Previous Vs Latest: $0.50 \leq x_F < 0.55$

## Previous

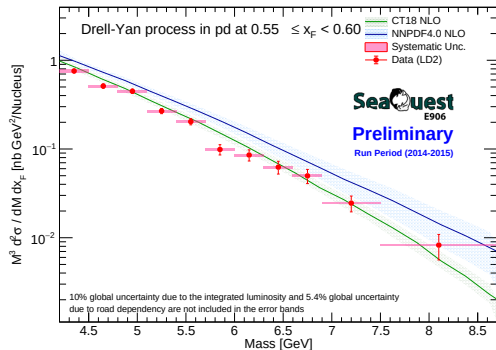


## Latest (with weighted average)

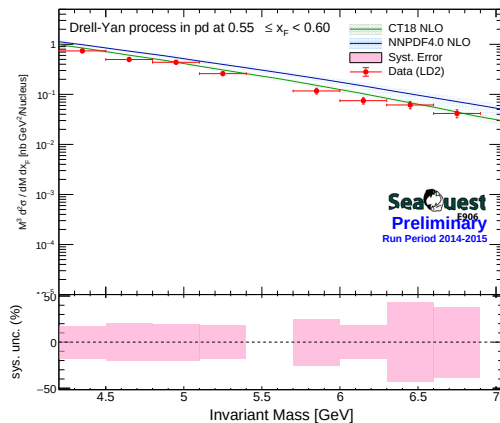


# LD2 Cross-Section Previous Vs Latest: $0.55 \leq x_F < 0.60$

## Previous

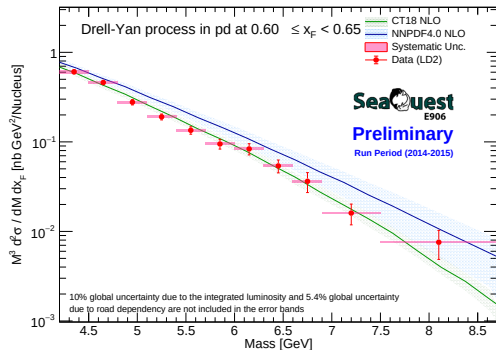


## Latest (with weighted average)

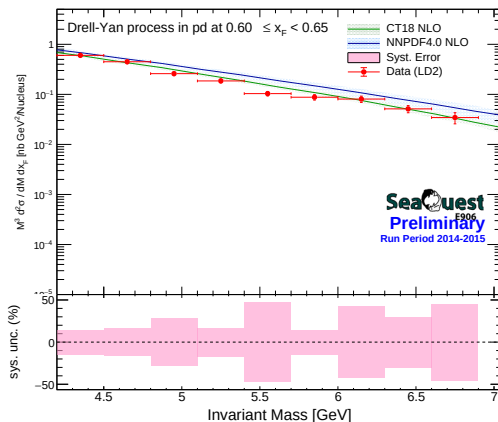


# LD2 Cross-Section Previous Vs Latest: $0.60 \leq x_F < 0.65$

## Previous

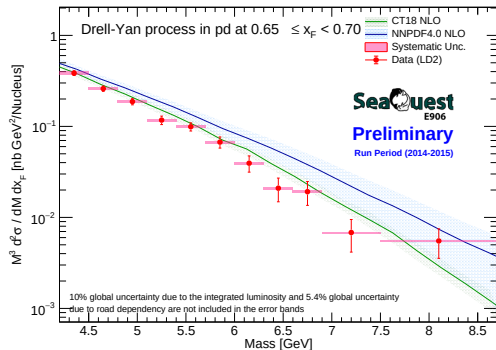


## Latest (with weighted average)

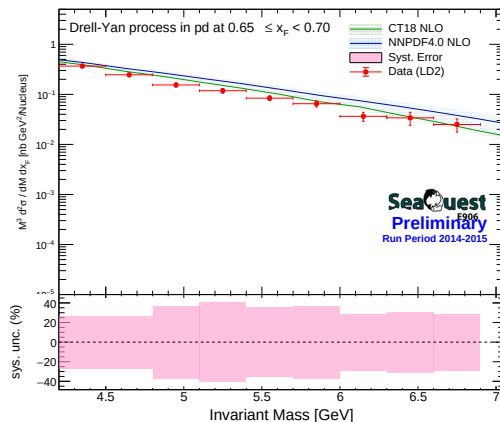


# LD2 Cross-Section Previous Vs Latest: $0.65 \leq x_F < 0.70$

## Previous

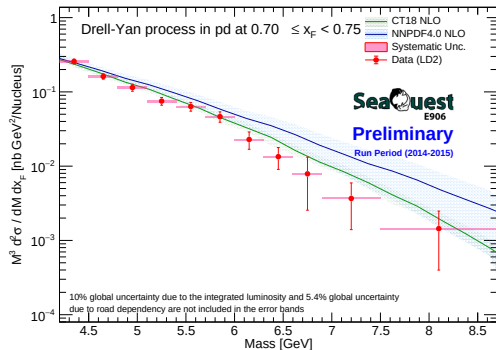


## Latest (with weighted average)

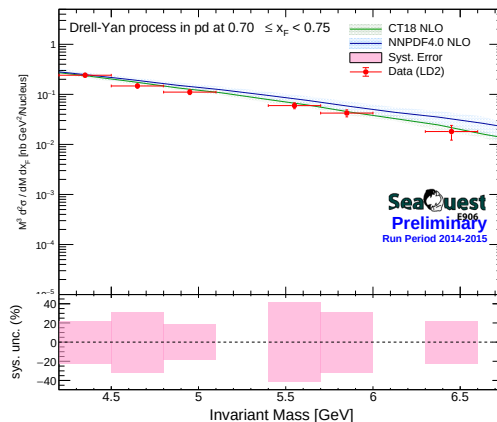


# LD2 Cross-Section Previous Vs Latest: $0.70 \leq x_F < 0.75$

## Previous

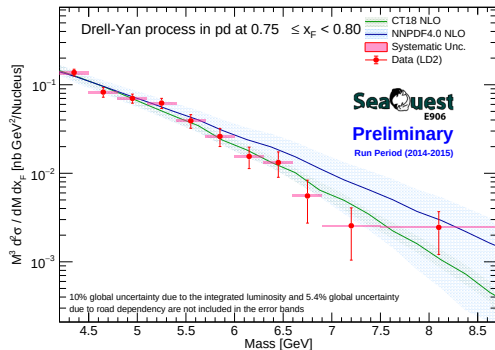


## Latest (with weighted average)

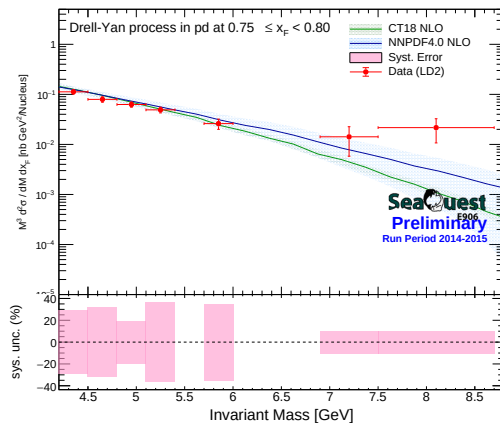


# LD2 Cross-Section Previous Vs Latest: $0.75 \leq x_F < 0.80$

## Previous

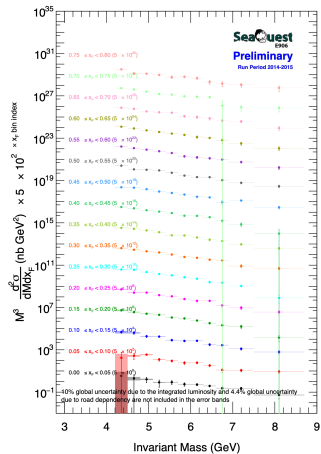


## Latest (with weighted average)

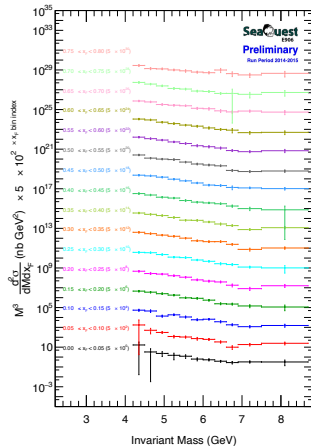


# Summary Plots Previous Vs Latest with updated systematics

## Previous

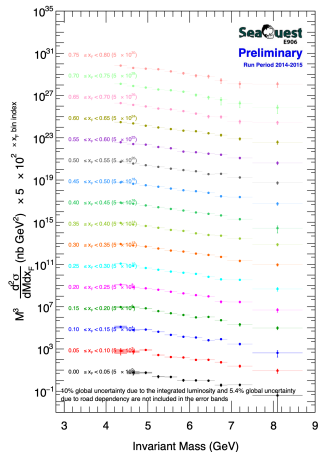


## Latest (with weighted average)

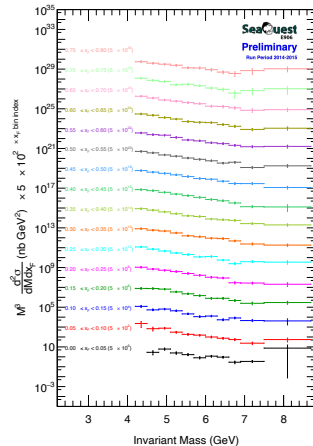


# Summary Plots Previous Vs Latest with updated systematics

## Previous



## Latest (with weighted average)



# Backup Slides

## Extension to 2D Kinematics:

- The inverse-variance weighting procedure is extended to the double differential bins ( $x_F$  and Invariant Mass).
- The following slides document the systematic uncertainties strictly for statistically significant data sets across both LH2 and LD2 targets.
- **Filtering Criteria:** Bins reporting a relative variance  $\leq 0\%$  or  $\geq 50\%$  represent zero-bin artifacts or severe low-statistics starvation and have been omitted for clarity.

# Double Differential Systematics (xF and Mass) - LH2 (1/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.00, 0.05) | [2.20, 2.30)     | 4.6647e-01      | 1.8959e-01               | 40.64%         |
| [0.00, 0.05) | [2.40, 2.50)     | 2.4243e-01      | 9.2653e-02               | 38.22%         |
| [0.00, 0.05) | [2.50, 2.60)     | 1.2537e-01      | 5.1962e-02               | 41.45%         |
| [0.00, 0.05) | [2.60, 2.70)     | 1.0592e-01      | 4.2068e-02               | 39.72%         |
| [0.05, 0.10) | [2.10, 2.20)     | 1.0106e+00      | 3.5192e-01               | 34.82%         |
| [0.05, 0.10) | [2.20, 2.30)     | 6.1206e-01      | 2.0965e-01               | 34.25%         |
| [0.05, 0.10) | [2.30, 2.40)     | 2.3946e-01      | 1.1893e-01               | 49.67%         |
| [0.05, 0.10) | [2.40, 2.50)     | 2.1213e-01      | 4.1157e-02               | 19.40%         |
| [0.05, 0.10) | [2.50, 2.60)     | 1.4685e-01      | 5.9468e-02               | 40.50%         |
| [0.05, 0.10) | [2.60, 2.70)     | 1.2250e-01      | 2.8580e-02               | 23.33%         |
| [0.05, 0.10) | [2.70, 2.80)     | 6.7852e-02      | 3.2529e-02               | 47.94%         |

## Double Differential Systematics (xF and Mass) - LH2 (2/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.10, 0.15) | [2.30, 2.40)     | 3.6968e-01      | 1.0765e-01               | 29.12%         |
| [0.10, 0.15) | [2.40, 2.50)     | 2.2354e-01      | 4.5725e-02               | 20.46%         |
| [0.10, 0.15) | [2.60, 2.70)     | 1.3294e-01      | 4.1845e-02               | 31.48%         |
| [0.10, 0.15) | [2.70, 2.80)     | 7.3985e-02      | 2.4278e-02               | 32.82%         |
| [0.15, 0.20) | [2.00, 2.10)     | 9.1591e-01      | 4.3322e-01               | 47.30%         |
| [0.15, 0.20) | [2.10, 2.20)     | 7.1875e-01      | 2.0705e-01               | 28.81%         |
| [0.15, 0.20) | [2.20, 2.30)     | 5.0253e-01      | 1.2457e-01               | 24.79%         |
| [0.15, 0.20) | [2.30, 2.40)     | 3.6176e-01      | 1.2212e-01               | 33.76%         |
| [0.15, 0.20) | [2.40, 2.50)     | 1.9252e-01      | 5.2335e-02               | 27.18%         |
| [0.15, 0.20) | [2.50, 2.60)     | 1.7226e-01      | 7.6151e-02               | 44.21%         |
| [0.15, 0.20) | [2.60, 2.70)     | 1.1220e-01      | 2.8566e-02               | 25.46%         |

# Double Differential Systematics (xF and Mass) - LH2 (3/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.20, 0.25) | [2.10, 2.20)     | 5.4657e-01      | 2.4949e-01               | 45.65%         |
| [0.20, 0.25) | [2.20, 2.30)     | 4.8145e-01      | 1.6689e-01               | 34.66%         |
| [0.20, 0.25) | [2.30, 2.40)     | 3.2918e-01      | 5.4812e-02               | 16.65%         |
| [0.20, 0.25) | [2.40, 2.50)     | 2.3267e-01      | 3.8954e-02               | 16.74%         |
| [0.20, 0.25) | [2.60, 2.70)     | 9.4423e-02      | 3.9979e-02               | 42.34%         |
| [0.20, 0.25) | [2.70, 2.80)     | 7.7348e-02      | 1.0475e-02               | 13.54%         |
| [0.20, 0.25) | [2.80, 2.90)     | 3.6159e-02      | 5.6716e-03               | 15.69%         |
| [0.25, 0.30) | [2.00, 2.10)     | 7.5260e-01      | 1.2825e-01               | 17.04%         |
| [0.25, 0.30) | [2.10, 2.20)     | 6.9524e-01      | 1.9496e-01               | 28.04%         |
| [0.25, 0.30) | [2.20, 2.30)     | 4.5304e-01      | 5.9077e-02               | 13.04%         |
| [0.25, 0.30) | [2.30, 2.40)     | 2.4852e-01      | 8.9635e-02               | 36.07%         |

# Double Differential Systematics (xF and Mass) - LH2 (4/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.25, 0.30) | [2.50, 2.60)     | 1.2740e-01      | 4.2815e-02               | 33.61%         |
| [0.25, 0.30) | [2.60, 2.70)     | 9.4024e-02      | 1.5640e-02               | 16.63%         |
| [0.25, 0.30) | [2.70, 2.80)     | 5.3786e-02      | 1.6526e-02               | 30.72%         |
| [0.25, 0.30) | [2.80, 2.90)     | 4.4295e-02      | 1.9575e-02               | 44.19%         |
| [0.25, 0.30) | [2.90, 3.00)     | 2.4688e-02      | 9.9960e-03               | 40.49%         |
| [0.30, 0.35) | [2.00, 2.10)     | 7.9903e-01      | 2.3076e-01               | 28.88%         |
| [0.30, 0.35) | [2.10, 2.20)     | 5.8222e-01      | 1.2269e-01               | 21.07%         |
| [0.30, 0.35) | [2.20, 2.30)     | 4.1542e-01      | 6.2699e-02               | 15.09%         |
| [0.30, 0.35) | [2.30, 2.40)     | 3.1708e-01      | 4.5316e-02               | 14.29%         |
| [0.30, 0.35) | [2.40, 2.50)     | 2.0555e-01      | 3.2558e-02               | 15.84%         |
| [0.30, 0.35) | [2.50, 2.60)     | 1.1104e-01      | 3.6713e-02               | 33.06%         |

# Double Differential Systematics (xF and Mass) - LH2 (5/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.30, 0.35) | [2.70, 2.80)     | 9.1170e-02      | 1.2310e-02               | 13.50%         |
| [0.30, 0.35) | [2.80, 2.90)     | 4.8748e-02      | 9.6601e-03               | 19.82%         |
| [0.35, 0.40) | [2.00, 2.10)     | 7.0876e-01      | 1.0139e-01               | 14.31%         |
| [0.35, 0.40) | [2.10, 2.20)     | 5.2199e-01      | 1.5482e-01               | 29.66%         |
| [0.35, 0.40) | [2.20, 2.30)     | 3.9245e-01      | 6.0602e-02               | 15.44%         |
| [0.35, 0.40) | [2.30, 2.40)     | 2.5106e-01      | 4.9200e-02               | 19.60%         |
| [0.35, 0.40) | [2.40, 2.50)     | 1.3409e-01      | 2.9509e-02               | 22.01%         |
| [0.35, 0.40) | [2.50, 2.60)     | 1.2855e-01      | 3.2383e-02               | 25.19%         |
| [0.35, 0.40) | [2.60, 2.70)     | 8.8306e-02      | 1.4040e-02               | 15.90%         |
| [0.35, 0.40) | [2.70, 2.80)     | 5.5295e-02      | 2.4767e-02               | 44.79%         |
| [0.35, 0.40) | [2.80, 2.90)     | 3.8024e-02      | 1.6029e-02               | 42.16%         |

# Double Differential Systematics (xF and Mass) - LH2 (6/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.40, 0.45) | [2.10, 2.20)     | 4.6687e-01      | 1.0521e-01               | 22.54%         |
| [0.40, 0.45) | [2.20, 2.30)     | 2.8022e-01      | 9.1991e-02               | 32.83%         |
| [0.40, 0.45) | [2.30, 2.40)     | 2.3071e-01      | 3.3997e-02               | 14.74%         |
| [0.40, 0.45) | [2.50, 2.60)     | 8.9501e-02      | 2.7328e-02               | 30.53%         |
| [0.40, 0.45) | [2.60, 2.70)     | 7.4179e-02      | 2.9933e-02               | 40.35%         |
| [0.40, 0.45) | [2.70, 2.80)     | 6.1148e-02      | 1.5443e-02               | 25.26%         |
| [0.40, 0.45) | [2.80, 2.90)     | 3.5909e-02      | 1.1889e-02               | 33.11%         |
| [0.45, 0.50) | [2.00, 2.10)     | 4.6946e-01      | 8.7180e-02               | 18.57%         |
| [0.45, 0.50) | [2.10, 2.20)     | 4.0052e-01      | 9.1889e-02               | 22.94%         |
| [0.45, 0.50) | [2.20, 2.30)     | 2.9739e-01      | 4.2426e-02               | 14.27%         |
| [0.45, 0.50) | [2.30, 2.40)     | 1.8520e-01      | 2.3280e-02               | 12.57%         |

# Double Differential Systematics (xF and Mass) - LH2 (7/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.45, 0.50) | [2.50, 2.60)     | 7.6792e-02      | 3.5737e-02               | 46.54%         |
| [0.45, 0.50) | [2.60, 2.70)     | 4.7282e-02      | 1.9263e-02               | 40.74%         |
| [0.50, 0.55) | [2.00, 2.10)     | 5.1561e-01      | 4.3113e-02               | 8.36%          |
| [0.50, 0.55) | [2.10, 2.20)     | 2.4825e-01      | 3.8514e-02               | 15.51%         |
| [0.50, 0.55) | [2.20, 2.30)     | 1.9861e-01      | 2.6949e-02               | 13.57%         |
| [0.50, 0.55) | [2.30, 2.40)     | 1.7728e-01      | 5.6211e-02               | 31.71%         |
| [0.50, 0.55) | [2.40, 2.50)     | 9.9353e-02      | 2.3469e-02               | 23.62%         |
| [0.50, 0.55) | [2.50, 2.60)     | 6.3380e-02      | 1.6129e-02               | 25.45%         |
| [0.50, 0.55) | [2.60, 2.70)     | 5.0313e-02      | 1.0344e-02               | 20.56%         |
| [0.50, 0.55) | [2.70, 2.80)     | 4.1483e-02      | 9.6799e-03               | 23.33%         |
| [0.55, 0.60) | [2.00, 2.10)     | 3.2536e-01      | 1.0254e-02               | 3.15%          |

# Double Differential Systematics (xF and Mass) - LH2 (8/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.55, 0.60) | [2.20, 2.30)     | 1.6549e-01      | 3.3759e-02               | 20.40%         |
| [0.55, 0.60) | [2.30, 2.40)     | 1.0744e-01      | 3.5247e-02               | 32.81%         |
| [0.55, 0.60) | [2.40, 2.50)     | 7.4276e-02      | 1.4861e-02               | 20.01%         |
| [0.55, 0.60) | [2.50, 2.60)     | 4.4520e-02      | 1.0677e-02               | 23.98%         |
| [0.55, 0.60) | [2.60, 2.70)     | 3.5292e-02      | 1.6495e-02               | 46.74%         |
| [0.55, 0.60) | [2.70, 2.80)     | 2.6122e-02      | 6.3284e-03               | 24.23%         |
| [0.60, 0.65) | [2.00, 2.10)     | 2.1866e-01      | 8.9135e-02               | 40.76%         |
| [0.60, 0.65) | [2.10, 2.20)     | 1.7900e-01      | 2.4785e-02               | 13.85%         |
| [0.60, 0.65) | [2.20, 2.30)     | 1.0609e-01      | 1.6341e-02               | 15.40%         |
| [0.60, 0.65) | [2.30, 2.40)     | 7.6175e-02      | 1.6733e-02               | 21.97%         |
| [0.60, 0.65) | [2.40, 2.50)     | 5.4076e-02      | 1.3899e-02               | 25.70%         |

# Double Differential Systematics (xF and Mass) - LH2 (9/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.60, 0.65) | [2.60, 2.70)     | 2.7941e-02      | 1.3177e-02               | 47.16%         |
| [0.65, 0.70) | [2.00, 2.10)     | 1.4650e-01      | 1.7599e-02               | 12.01%         |
| [0.65, 0.70) | [2.10, 2.20)     | 1.1789e-01      | 1.4840e-02               | 12.59%         |
| [0.65, 0.70) | [2.20, 2.30)     | 6.4554e-02      | 1.7128e-02               | 26.53%         |
| [0.65, 0.70) | [2.30, 2.40)     | 5.4773e-02      | 2.1213e-02               | 38.73%         |
| [0.65, 0.70) | [2.40, 2.50)     | 4.8254e-02      | 8.8236e-03               | 18.29%         |
| [0.65, 0.70) | [2.50, 2.60)     | 2.6751e-02      | 1.1587e-02               | 43.31%         |
| [0.65, 0.70) | [2.60, 2.70)     | 2.4311e-02      | 5.1266e-03               | 21.09%         |
| [0.70, 0.75) | [2.00, 2.10)     | 1.1002e-01      | 1.0678e-02               | 9.71%          |
| [0.70, 0.75) | [2.10, 2.20)     | 7.4866e-02      | 2.2381e-02               | 29.89%         |
| [0.70, 0.75) | [2.20, 2.30)     | 4.8945e-02      | 1.2593e-02               | 25.73%         |

## Double Differential Systematics (xF and Mass) - LH2 (10/10)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.70, 0.75) | [2.40, 2.50)     | 2.3570e-02      | 8.6238e-03               | 36.59%         |
| [0.70, 0.75) | [2.60, 2.70)     | 1.3325e-02      | 5.8748e-03               | 44.09%         |
| [0.75, 0.80) | [2.00, 2.10)     | 5.6607e-02      | 1.2918e-02               | 22.82%         |
| [0.75, 0.80) | [2.20, 2.30)     | 2.5546e-02      | 1.0505e-02               | 41.12%         |
| [0.75, 0.80) | [2.30, 2.40)     | 2.0004e-02      | 8.5603e-03               | 42.79%         |
| [0.75, 0.80) | [2.40, 2.50)     | 1.6363e-02      | 6.0294e-03               | 36.85%         |
| [0.75, 0.80) | [2.50, 2.60)     | 1.2196e-02      | 6.0025e-03               | 49.21%         |

# Double Differential Systematics (xF and Mass) - LD2 (1/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.00, 0.05) | [2.20, 2.30)     | 1.2134e+00      | 3.5467e-01               | 29.23%         |
| [0.00, 0.05) | [2.30, 2.40)     | 4.9300e-01      | 2.2986e-01               | 46.62%         |
| [0.00, 0.05) | [2.60, 2.70)     | 2.3350e-01      | 7.9457e-02               | 34.03%         |
| [0.00, 0.05) | [2.70, 2.80)     | 1.8693e-01      | 5.7193e-02               | 30.60%         |
| [0.05, 0.10) | [2.20, 2.30)     | 1.5352e+00      | 2.7104e-01               | 17.66%         |
| [0.05, 0.10) | [2.30, 2.40)     | 6.1204e-01      | 1.1169e-01               | 18.25%         |
| [0.05, 0.10) | [2.40, 2.50)     | 3.6952e-01      | 1.6648e-01               | 45.05%         |
| [0.05, 0.10) | [2.50, 2.60)     | 3.0688e-01      | 1.1042e-01               | 35.98%         |
| [0.05, 0.10) | [2.60, 2.70)     | 1.9908e-01      | 3.1577e-02               | 15.86%         |
| [0.05, 0.10) | [2.80, 2.90)     | 1.0977e-01      | 4.3818e-02               | 39.92%         |
| [0.10, 0.15) | [2.00, 2.10)     | 2.4320e+00      | 4.6782e-01               | 19.24%         |

# Double Differential Systematics (xF and Mass) - LD2 (2/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.10, 0.15) | [2.30, 2.40)     | 8.7415e-01      | 1.3958e-01               | 15.97%         |
| [0.10, 0.15) | [2.40, 2.50)     | 4.2530e-01      | 5.9980e-02               | 14.10%         |
| [0.10, 0.15) | [2.50, 2.60)     | 2.2473e-01      | 7.1942e-02               | 32.01%         |
| [0.10, 0.15) | [2.60, 2.70)     | 2.5785e-01      | 5.2115e-02               | 20.21%         |
| [0.10, 0.15) | [2.80, 2.90)     | 1.6552e-01      | 6.1436e-02               | 37.12%         |
| [0.15, 0.20) | [2.00, 2.10)     | 1.5687e+00      | 6.1198e-01               | 39.01%         |
| [0.15, 0.20) | [2.10, 2.20)     | 1.5232e+00      | 7.4036e-01               | 48.60%         |
| [0.15, 0.20) | [2.20, 2.30)     | 1.3359e+00      | 1.9766e-01               | 14.80%         |
| [0.15, 0.20) | [2.30, 2.40)     | 7.0016e-01      | 2.3754e-01               | 33.93%         |
| [0.15, 0.20) | [2.40, 2.50)     | 4.4981e-01      | 8.3544e-02               | 18.57%         |
| [0.15, 0.20) | [2.50, 2.60)     | 2.9692e-01      | 1.4121e-01               | 47.56%         |

# Double Differential Systematics (xF and Mass) - LD2 (3/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.15, 0.20) | [2.70, 2.80)     | 1.5843e-01      | 7.3299e-02               | 46.27%         |
| [0.15, 0.20) | [2.80, 2.90)     | 9.5246e-02      | 2.3443e-02               | 24.61%         |
| [0.20, 0.25) | [2.00, 2.10)     | 2.1372e+00      | 3.8561e-01               | 18.04%         |
| [0.20, 0.25) | [2.10, 2.20)     | 1.3985e+00      | 1.9926e-01               | 14.25%         |
| [0.20, 0.25) | [2.20, 2.30)     | 1.0683e+00      | 3.9122e-01               | 36.62%         |
| [0.20, 0.25) | [2.30, 2.40)     | 7.8453e-01      | 1.8395e-01               | 23.45%         |
| [0.20, 0.25) | [2.40, 2.50)     | 5.0149e-01      | 1.1740e-01               | 23.41%         |
| [0.20, 0.25) | [2.50, 2.60)     | 3.3169e-01      | 1.2986e-01               | 39.15%         |
| [0.20, 0.25) | [2.60, 2.70)     | 2.1104e-01      | 5.5104e-02               | 26.11%         |
| [0.20, 0.25) | [2.70, 2.80)     | 1.7673e-01      | 7.7991e-02               | 44.13%         |
| [0.20, 0.25) | [2.80, 2.90)     | 5.8350e-02      | 2.8937e-02               | 49.59%         |

# Double Differential Systematics (xF and Mass) - LD2 (4/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.25, 0.30) | [2.10, 2.20)     | 1.4279e+00      | 2.9626e-01               | 20.75%         |
| [0.25, 0.30) | [2.20, 2.30)     | 9.2956e-01      | 1.5364e-01               | 16.53%         |
| [0.25, 0.30) | [2.30, 2.40)     | 5.9333e-01      | 1.2729e-01               | 21.45%         |
| [0.25, 0.30) | [2.40, 2.50)     | 4.0752e-01      | 1.1414e-01               | 28.01%         |
| [0.25, 0.30) | [2.60, 2.70)     | 2.5680e-01      | 4.2894e-02               | 16.70%         |
| [0.25, 0.30) | [2.70, 2.80)     | 1.1377e-01      | 3.9897e-02               | 35.07%         |
| [0.25, 0.30) | [2.90, 3.00)     | 7.7665e-02      | 2.5707e-02               | 33.10%         |
| [0.30, 0.35) | [2.00, 2.10)     | 1.6374e+00      | 4.2036e-01               | 25.67%         |
| [0.30, 0.35) | [2.10, 2.20)     | 1.1762e+00      | 3.4070e-01               | 28.97%         |
| [0.30, 0.35) | [2.20, 2.30)     | 9.6178e-01      | 1.4852e-01               | 15.44%         |
| [0.30, 0.35) | [2.30, 2.40)     | 6.1483e-01      | 1.5873e-01               | 25.82%         |

# Double Differential Systematics (xF and Mass) - LD2 (5/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.30, 0.35) | [2.70, 2.80)     | 1.6502e-01      | 4.4954e-02               | 27.24%         |
| [0.30, 0.35) | [2.80, 2.90)     | 9.6576e-02      | 3.8679e-02               | 40.05%         |
| [0.30, 0.35) | [2.90, 3.00)     | 4.7350e-02      | 1.9646e-02               | 41.49%         |
| [0.35, 0.40) | [2.00, 2.10)     | 1.7182e+00      | 1.1746e-01               | 6.84%          |
| [0.35, 0.40) | [2.10, 2.20)     | 1.2202e+00      | 1.6988e-01               | 13.92%         |
| [0.35, 0.40) | [2.20, 2.30)     | 8.1179e-01      | 2.7942e-01               | 34.42%         |
| [0.35, 0.40) | [2.30, 2.40)     | 5.2692e-01      | 2.1254e-01               | 40.34%         |
| [0.35, 0.40) | [2.40, 2.50)     | 3.2734e-01      | 1.3486e-01               | 41.20%         |
| [0.35, 0.40) | [2.50, 2.60)     | 2.7822e-01      | 8.7961e-02               | 31.62%         |
| [0.35, 0.40) | [2.60, 2.70)     | 1.5650e-01      | 4.5772e-02               | 29.25%         |
| [0.35, 0.40) | [2.80, 2.90)     | 1.1346e-01      | 1.5148e-02               | 13.35%         |

# Double Differential Systematics (xF and Mass) - LD2 (6/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.40, 0.45) | [2.10, 2.20)     | 1.1311e+00      | 7.9911e-02               | 7.07%          |
| [0.40, 0.45) | [2.20, 2.30)     | 7.6144e-01      | 1.3688e-01               | 17.98%         |
| [0.40, 0.45) | [2.30, 2.40)     | 5.0685e-01      | 7.9701e-02               | 15.72%         |
| [0.40, 0.45) | [2.40, 2.50)     | 3.5430e-01      | 5.2289e-02               | 14.76%         |
| [0.40, 0.45) | [2.50, 2.60)     | 2.3511e-01      | 6.0446e-02               | 25.71%         |
| [0.40, 0.45) | [2.60, 2.70)     | 1.4468e-01      | 5.7135e-02               | 39.49%         |
| [0.40, 0.45) | [2.70, 2.80)     | 1.0666e-01      | 4.9775e-02               | 46.67%         |
| [0.40, 0.45) | [2.80, 2.90)     | 6.4523e-02      | 2.8254e-02               | 43.79%         |
| [0.45, 0.50) | [2.00, 2.10)     | 1.1839e+00      | 1.6772e-01               | 14.17%         |
| [0.45, 0.50) | [2.10, 2.20)     | 8.2842e-01      | 2.0026e-01               | 24.17%         |
| [0.45, 0.50) | [2.20, 2.30)     | 6.2343e-01      | 1.4134e-01               | 22.67%         |

# Double Differential Systematics (xF and Mass) - LD2 (7/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.45, 0.50) | [2.40, 2.50)     | 2.9707e-01      | 5.0997e-02               | 17.17%         |
| [0.45, 0.50) | [2.50, 2.60)     | 1.8791e-01      | 7.8155e-02               | 41.59%         |
| [0.45, 0.50) | [2.60, 2.70)     | 1.5975e-01      | 1.9869e-02               | 12.44%         |
| [0.45, 0.50) | [2.70, 2.80)     | 1.0093e-01      | 4.0190e-02               | 39.82%         |
| [0.45, 0.50) | [2.90, 3.00)     | 5.7965e-02      | 2.3706e-02               | 40.90%         |
| [0.50, 0.55) | [2.00, 2.10)     | 1.0189e+00      | 1.7374e-01               | 17.05%         |
| [0.50, 0.55) | [2.10, 2.20)     | 7.2468e-01      | 4.5340e-02               | 6.26%          |
| [0.50, 0.55) | [2.20, 2.30)     | 4.4702e-01      | 1.4831e-01               | 33.18%         |
| [0.50, 0.55) | [2.30, 2.40)     | 3.3354e-01      | 7.6521e-02               | 22.94%         |
| [0.50, 0.55) | [2.40, 2.50)     | 2.1909e-01      | 6.4012e-02               | 29.22%         |
| [0.50, 0.55) | [2.50, 2.60)     | 1.6166e-01      | 5.5508e-02               | 34.34%         |

# Double Differential Systematics (xF and Mass) - LD2 (8/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.50, 0.55) | [2.70, 2.80)     | 6.7888e-02      | 7.7766e-03               | 11.45%         |
| [0.50, 0.55) | [2.80, 2.90)     | 8.2012e-02      | 1.7043e-02               | 20.78%         |
| [0.55, 0.60) | [2.00, 2.10)     | 7.3861e-01      | 1.0244e-01               | 13.87%         |
| [0.55, 0.60) | [2.10, 2.20)     | 4.9734e-01      | 8.5445e-02               | 17.18%         |
| [0.55, 0.60) | [2.20, 2.30)     | 4.3767e-01      | 7.4056e-02               | 16.92%         |
| [0.55, 0.60) | [2.30, 2.40)     | 2.5958e-01      | 3.8744e-02               | 14.93%         |
| [0.55, 0.60) | [2.50, 2.60)     | 1.1703e-01      | 2.7149e-02               | 23.20%         |
| [0.55, 0.60) | [2.60, 2.70)     | 7.4815e-02      | 1.1092e-02               | 14.83%         |
| [0.55, 0.60) | [2.70, 2.80)     | 6.1416e-02      | 2.5615e-02               | 41.71%         |
| [0.55, 0.60) | [2.80, 2.90)     | 4.1506e-02      | 1.5355e-02               | 37.00%         |
| [0.60, 0.65) | [2.00, 2.10)     | 6.0184e-01      | 5.6361e-02               | 9.36%          |

# Double Differential Systematics (xF and Mass) - LD2 (9/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.60, 0.65) | [2.20, 2.30)     | 2.5929e-01      | 6.7410e-02               | 26.00%         |
| [0.60, 0.65) | [2.30, 2.40)     | 1.8560e-01      | 2.5284e-02               | 13.62%         |
| [0.60, 0.65) | [2.40, 2.50)     | 1.0353e-01      | 4.7449e-02               | 45.83%         |
| [0.60, 0.65) | [2.50, 2.60)     | 8.7706e-02      | 8.9969e-03               | 10.26%         |
| [0.60, 0.65) | [2.60, 2.70)     | 8.0616e-02      | 3.3267e-02               | 41.27%         |
| [0.60, 0.65) | [2.70, 2.80)     | 5.1333e-02      | 1.4382e-02               | 28.02%         |
| [0.60, 0.65) | [2.80, 2.90)     | 3.4400e-02      | 1.5218e-02               | 44.24%         |
| [0.65, 0.70) | [2.00, 2.10)     | 3.6757e-01      | 8.9009e-02               | 24.22%         |
| [0.65, 0.70) | [2.10, 2.20)     | 2.4621e-01      | 6.0907e-02               | 24.74%         |
| [0.65, 0.70) | [2.20, 2.30)     | 1.5429e-01      | 5.4661e-02               | 35.43%         |
| [0.65, 0.70) | [2.30, 2.40)     | 1.1833e-01      | 4.6189e-02               | 39.03%         |

# Double Differential Systematics (xF and Mass) - LD2 (10/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.65, 0.70) | [2.50, 2.60)     | 6.5078e-02      | 2.3231e-02               | 35.70%         |
| [0.65, 0.70) | [2.60, 2.70)     | 3.6397e-02      | 9.6883e-03               | 26.62%         |
| [0.65, 0.70) | [2.70, 2.80)     | 3.3985e-02      | 9.8990e-03               | 29.13%         |
| [0.65, 0.70) | [2.80, 2.90)     | 2.5011e-02      | 6.7470e-03               | 26.98%         |
| [0.70, 0.75) | [2.00, 2.10)     | 2.4122e-01      | 4.5913e-02               | 19.03%         |
| [0.70, 0.75) | [2.10, 2.20)     | 1.4718e-01      | 4.3820e-02               | 29.77%         |
| [0.70, 0.75) | [2.20, 2.30)     | 1.1097e-01      | 1.6766e-02               | 15.11%         |
| [0.70, 0.75) | [2.40, 2.50)     | 5.9655e-02      | 2.3853e-02               | 39.99%         |
| [0.70, 0.75) | [2.50, 2.60)     | 4.2258e-02      | 1.2634e-02               | 29.90%         |
| [0.70, 0.75) | [2.70, 2.80)     | 1.8082e-02      | 3.5747e-03               | 19.77%         |
| [0.75, 0.80) | [2.00, 2.10)     | 1.1242e-01      | 3.0062e-02               | 26.74%         |

## Double Differential Systematics (xF and Mass) - LD2 (11/11)

| xF Range     | Mass Range [GeV] | Mean $\sigma_w$ | Std Dev $\sigma_{sys,w}$ | Relative Error |
|--------------|------------------|-----------------|--------------------------|----------------|
| [0.75, 0.80) | [2.20, 2.30)     | 6.2737e-02      | 1.0170e-02               | 16.21%         |
| [0.75, 0.80) | [2.30, 2.40)     | 4.9022e-02      | 1.7038e-02               | 34.76%         |
| [0.75, 0.80) | [2.50, 2.60)     | 2.6054e-02      | 8.7078e-03               | 33.42%         |

## Why is the relative error still $\sim 13\text{--}18\%$ ?

- 1 **Statistical Noise Limitations:** While inverse-variance weighting heavily suppresses the noise of low-POT datasets (RS59, RS70), it cannot completely eliminate the pull from extreme outliers.
- 2 **Global Acceptance Mismatch:** A single `acceptance_mass_xF.root` file is currently used across all roadsets. Because different triggers have distinct kinematic acceptances, dividing by a global average creates artificial variance.

**Path Forward:** Implement roadset-specific Monte Carlo acceptance maps and evaluate the necessity of excluding extremely low-POT datasets from the overall variance calculation.